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Network Deployment and Alternative OSs

These are exciting times for devices. End users have the options of Android, ChromeOS, FirefoxOS, KDE Plasma Active, etc, to choose from. Here's a look at how similar or different these OSs are from each other.

One striking difference is that each OS is supposed to have a distinct personality in addition to, perhaps, some differences in functionality.

This is in contrast to the flexibility of distributions like Fedora and Arch Linux, where you can get very different environments by switching between KDE, GNOME, Cinnamon, XFCE, LXDE, etc. Ubuntu may have a preference for Unity, but it too can be changed to any of the other personalities.

You may also differentiate various environments on the basis of thin and thick clients. The new environments seem to fall between the two. They rely predominantly on the Web or the cloud, but do have local applications, which tend to be lightweight. They also rely upon a rich browser, which can work with local data as well. So, to use comparatively ancient terminology, would you consider these devices the 'clients' in a 'client-server' environment?

Given so many new choices, especially as developers, you need a way to play around with the various alternatives, and understand their specialities and idiosyncrasies. In a corporate or educational environment, you may need to support and manage the variations.

It is good to re-look at some of the old and not-so-old solutions, and see how they perform today. One of the ideas would be to check out if you could extend some of these solutions in order to incorporate the desirable features of the new environments in corporate and educational environments.

Running Linux from RAM

If you haven't tried Puppy Linux in a while, do try it. It is not that it is very different from what it was some years ago; but if you run it on a recent desktop, the user experience is remarkable. It is incredibly responsive.

Puppy Linux has packaged a limited set of applications, which are small in size. The default operation is that it runs from RAM. So, when you boot into Puppy Linux, it seems to take a comparatively long time for such a compact distribution; however, this time is needed to copy and set up the root partition in RAM.

Base Puppy Linux consists of three files—*vmlinuz*, *initrd.gz* and a root squash file system (something like *puppy_precise_5.4.3.sfs*). You may copy the three files in, say, a *puppy* sub-directory under the */boot* partition, and add an entry in the GRUB menu as follows:

```
menuentry 'Puppy Linux'{
    linux /boot/puppy/vmlinuz-puppy psudir=boot/puppy
    initrd /boot/puppy/initrd-puppy.gz
}
```

You may wish to experiment with booting from a server over the network. Since the Puppy Linux kernel does not come with NFS support, the easiest option is to create a new, larger *initrd* file by adding the *sfs* file in the *initrd*. You will need to suitably configure the *dhcpcd*, *tftp* and HTTP servers. The latter is important, as booting with *tftp* is slow for large files and iPXE boot ROMs (*ipxe.org*) support a variety of options, including HTTP. You may use *ipxe.krn* from <http://boot.ipxe.org/ipxe.iso>, which includes all the supported drivers. The GRUB entry is simple:

```
menuentry "ipxe"{
    kernel16 ipxe.krn
}
```

A comparison of booting times on the same hardware follows:

Linux Flavour	Boot-up Time
Ubuntu 12.10 from local disk	45 sec
Precise Puppy 5.4.3 from local disk	Almost 40 sec
Precise Puppy 5.4.3 using HTTP over 100 MBps net connection	Just over 1 min
Precise Puppy 5.4.3 using HTTP over 1 Gigabit net connection	50 sec